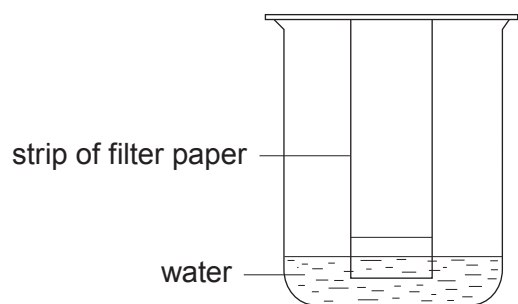
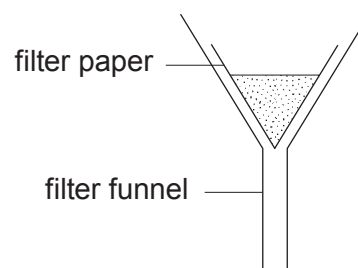


*Answer all questions.*

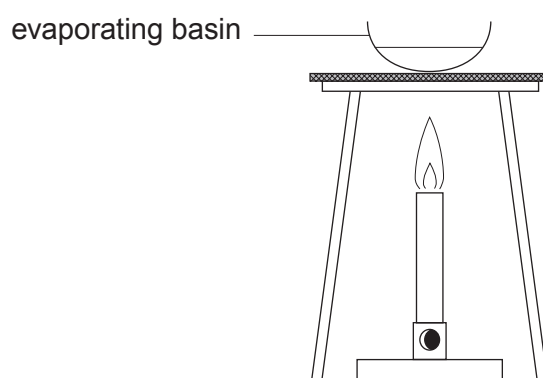
1. (a) The diagrams show three methods, **A**, **B** and **C**, used to separate mixtures.



**A**



**B**



**C**

Give the letter, **A**, **B** or **C**, of the method you would use to:

- (i) remove sand from sea-water, .....

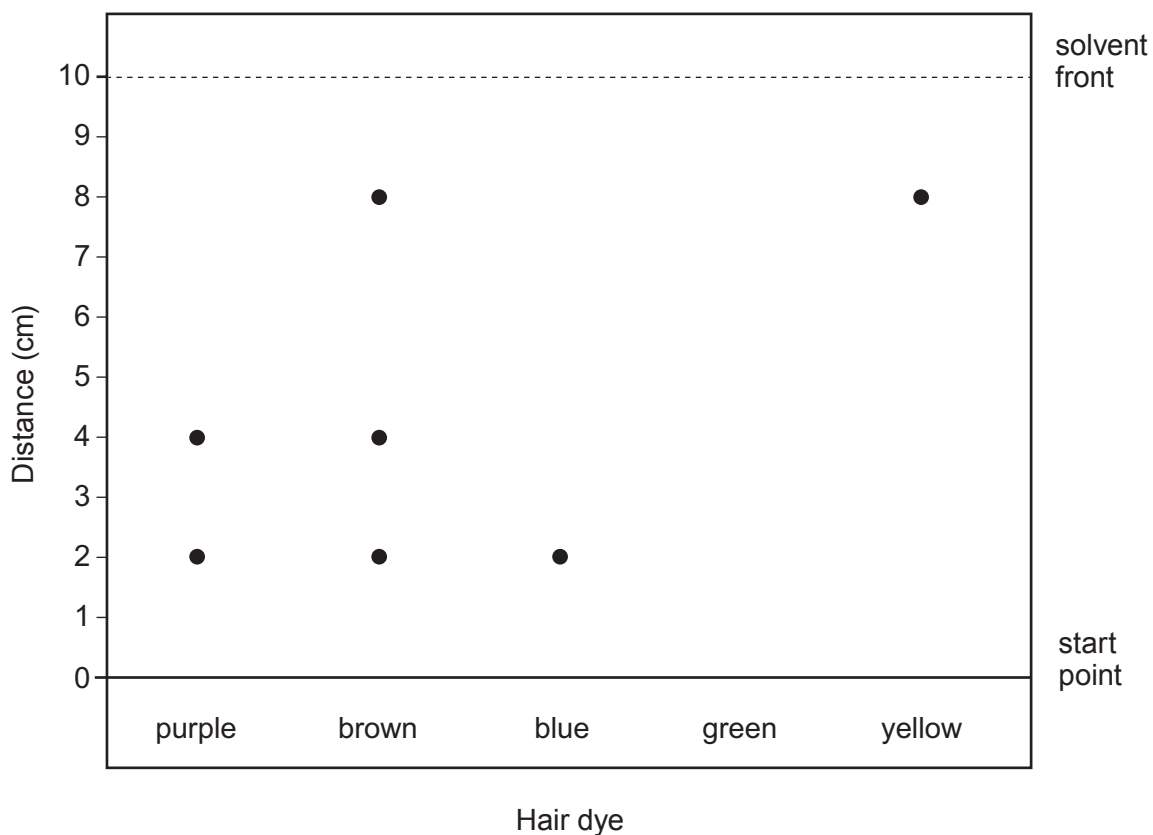
[1]

- (ii) show that sea-water contains salt. ....

[1]



(b) Paper chromatography was used to show the pigments present in different hair dyes.



(i) Give the **two** dyes that are mixed to make brown hair dye. [1]

..... and .....

(ii) Green hair dye is made by mixing together blue and yellow dyes.  
**Draw on the diagram** the result that you would expect to see for green hair dye. [1]

(iii) The  $R_f$  value of a substance can be used to identify that substance.  
The  $R_f$  value of the pigment in red hair dye is 0.4.

Use the equation below to calculate the distance this red hair dye would have moved on the diagram. [2]

$$\text{distance moved} = R_f \times \text{distance moved by solvent}$$

distance moved = ..... cm

6

